Z-transform and Inverse Z transform

- The Z-transform of a discrete time signal $x(n)$ is represented by $X(z)$

$$x(n) \xleftrightarrow{ZT} X(z)$$

- $$\boxed{X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}}$$

- $x(t) \longrightarrow$ freq $\longrightarrow$ F.T., L.T.

  $x(n) \longrightarrow$ freq $\longrightarrow$ D.F.T, Z.T

$n=-\infty$

- Relation bet.^{n} DFT & Z.T.

- $$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}$$

$$\boxed{z = r e^{j\omega}}$$

- $$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, (r e^{j\omega})^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x(n)\, r^{-n} e^{-j\omega n}$$

$$\mathrm{DFT}[x(n)] = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n}$$

- $$F.T.[x(t)] = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

- $$D.F.T.[x(n)] = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n}$$

- Relation betⁿ DFT & Z.T.

- $$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}$$

$$\boxed{z = re^{j\omega}}$$

- $$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, (re^{j\omega})^{-n}$$

$$= \sum_{n=-\infty}^{\infty} x(n)\, r^{-n} e^{-j\omega n}$$

$$DFT[x(n)] = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n}$$

- $\text{D.F.T.}[x(n)] = \sum_{n=-\infty} x(n)\, e$

- Relation bet.^{n} DFT & Z.T.

- $X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}$

$$\boxed{z = re^{j\omega}}$$

- $X(z) = \sum_{n=-\infty}^{\infty} x(n)\, (re^{j\omega})^{-n}$

$= \sum_{n=-\infty}^{\infty} \underline{x(n)\, r^{-n}}\, \underline{e}^{-j\omega n}$

$= \text{DFT}[x(n)\, r^{-n}]$

$\text{DFT}[x(n)] = \sum_{n=-\infty}^{\infty} x(n)\, \underline{e}^{-j\omega n}$

- $$X(z) = \sum_{n=-\infty}^{\infty} x(n) \, (r e^{j\omega})^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \underline{x(n) \, r^{-n}} \, \underline{e}^{-j\omega n}$$

$$= DFT \left[x(n) \, r^{-n} \right]$$

$$DFT[x(n)] = \sum_{n=-\infty}^{\infty} x(n) \, \underline{e}^{-j\omega n}$$

- Inverse Z - transform

$$\Rightarrow X(z) = DFT \left[x(n) \, r^{-n} \right]$$

$$= \sum_{n=-\infty}^{\infty} x(n)\, r^{-n}\, e^{-j\omega n}$$

$n=-\infty$

$$= DFT\left[x(n)\, r^{-n}\right]$$

- Inverse Z - transform

$$\Rightarrow X(z) = DFT\left[x(n)\, r^{-n}\right]$$

$$\Rightarrow IDFT\left[X(z)\right] = x(n)\, r^{-n}$$

$$\Rightarrow x(n) = r^{n}\left[IDFT\left(X(z)\right)\right]$$

$$= r^{n}\left[\frac{1}{2\pi}\int X(z)\, e^{j\omega n}\, d\omega\right]$$

$$= \frac{1}{2\pi}\int X(z)\left(r e^{j\omega}\right)^{n} d\omega$$

$\rightarrow z = r e^{j\omega}$

$$= \sum_{n=-\infty}^{\infty} \underline{x(n)\, r^{-n}}\, \underline{e^{-j\omega n}}$$

$n = -\infty$

$$= DFT\,[x(n)\, r^{-n}]$$

Inverse Z - transform

$$\Rightarrow X(z) = DFT\,[x(n)\, r^{-n}]$$

$$\Rightarrow IDFT\,[X(z)] = x(n)\, r^{-n}$$

$$\Rightarrow x(n) = r^{n}\,[\,IDFT\,(X(z))\,]$$

$$= r^{n}\left[\frac{1}{2\pi}\int X(z)\, e^{j\omega n}\, d\omega\right]$$

$$= \frac{1}{2\pi}\int X(z)\,(r e^{j\omega})^{n}\, d\omega$$

$$\rightarrow z = r e^{j\omega}$$

$$\Rightarrow z = r e^{j\omega}$$

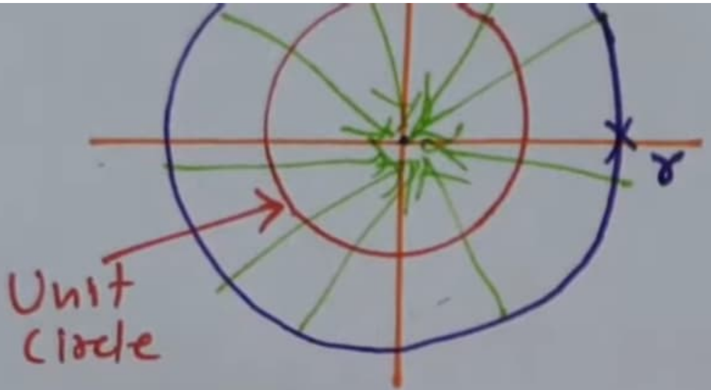
$$\Rightarrow dz = r e^{j\omega}\, j\, d\omega$$

$$\Rightarrow dz = jz\, d\omega$$

$$\Rightarrow d\omega = \frac{dz}{jz}$$

$$\Rightarrow x(n) = \frac{1}{2\pi}\int X(z)\, z^{n}\, \frac{dz}{jz}$$

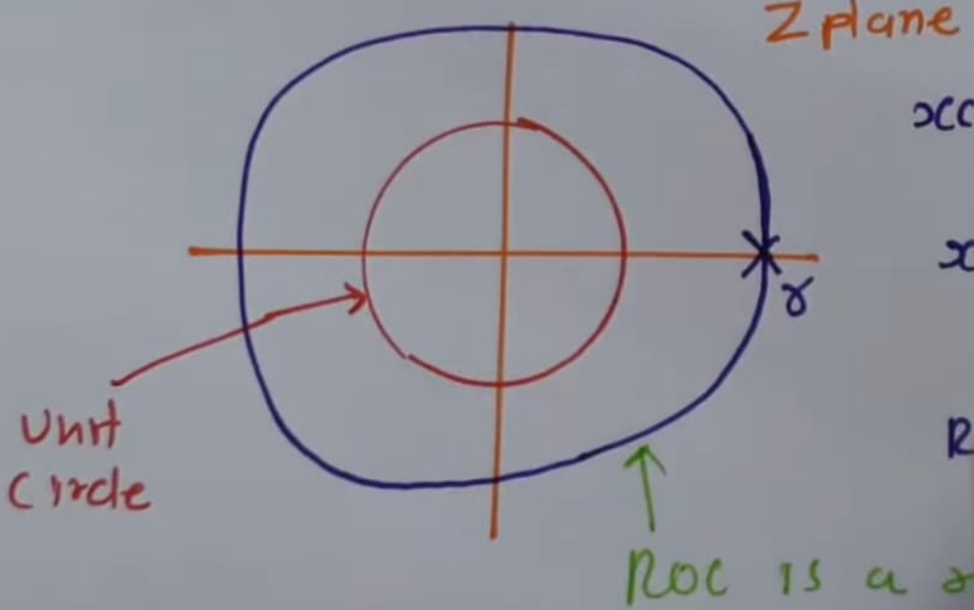
$$\Rightarrow \boxed{x(n) = \frac{1}{2\pi j}\int X(z)\, z^{n-1}\, dz}$$



$x(n) \longleftrightarrow X(z) = \frac{z}{z-\gamma}$

ROC : $|z| < \gamma$

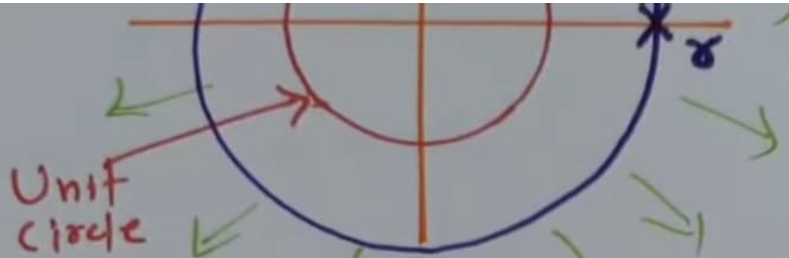
- If $x(n)$ is two sided signal & If $|z| = \gamma$ Circle is in ROC, Then the ROC will contain a ring in Z plane that include $|z| = \gamma$



$x(n) = f_1 u(n) + f_2 u(-n)$

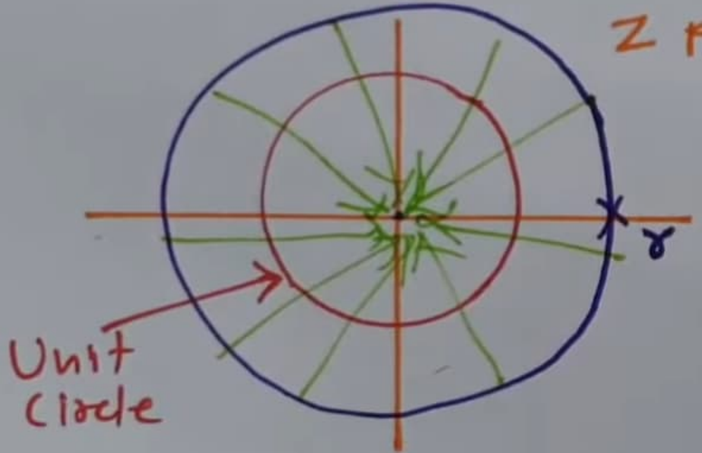
$x(n) \longleftrightarrow$ ... $= \frac{z}{z-\gamma}$

R ... $|z|$



ROC : $|z| > r$

- If $x(n)$ is Left sided sequence & if $|z| = r$ in the ROC, then all finite value of z, for which $|z| < r$ will also be in ROC



$$x(n) = f(n)\, \underline{\underline{u(-n)}}$$

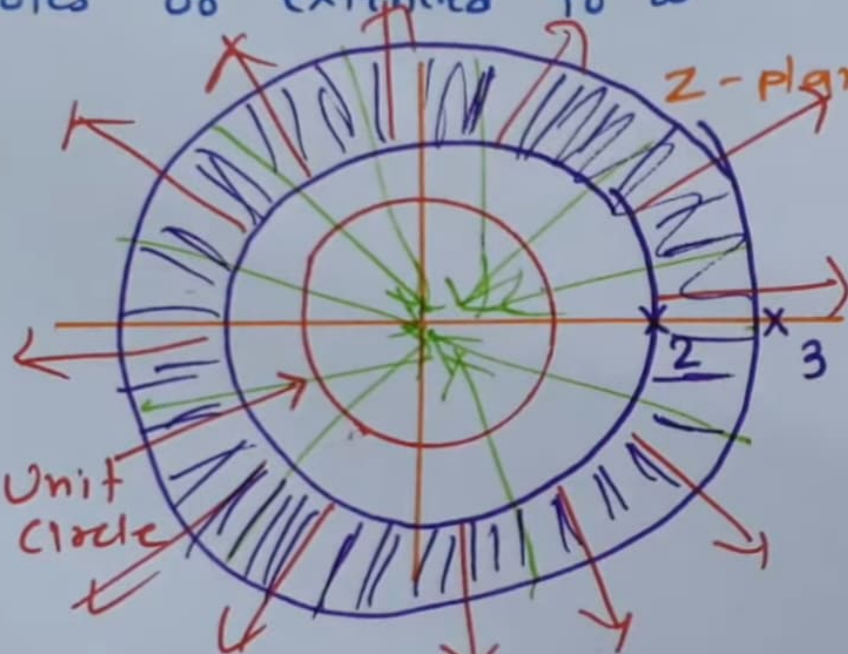
$$x(n) \longleftrightarrow X(z) = \frac{z}{z-r}$$

ROC : $|z| < r$

- If $x(n)$ is two sided signal & if $|z| = r$ circle is in ROC, Then the ROC will contain a ring in Z plane that include $|z| = r$

$x(n)$ … $u(-n)$

- If the z-Transform of $x(n)$ is rational, then its ROC is bounded by poles or extended to ∞



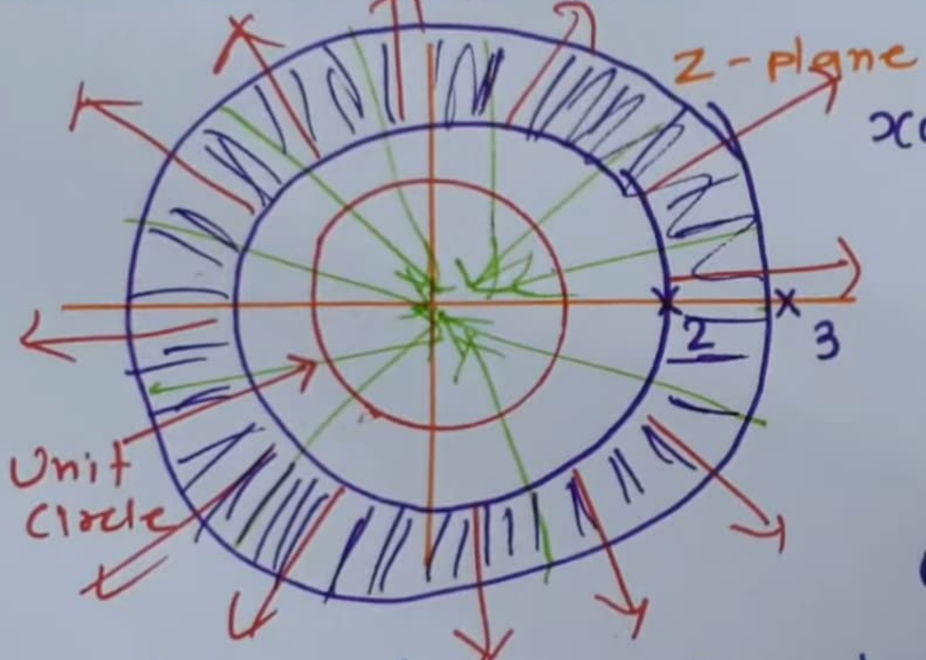
$$x(n) = 2^n u(n) - 3^n u(-n-1)$$

$$\frac{z}{z-2} \qquad \frac{z}{z-3}$$

$$|z| > 2 \qquad |z| < 3$$

- If the z-transform of $x(n)$ is rational & ... then ROC is z-plane outside the outer most ...

- If the Z-Transform of $x(n)$ is rational, then its ROC is bounded by poles or extended to ∞



$$x(n) = 2^n u(n) - 3^n u(-n-1)$$

$$\downarrow \qquad\qquad \downarrow$$

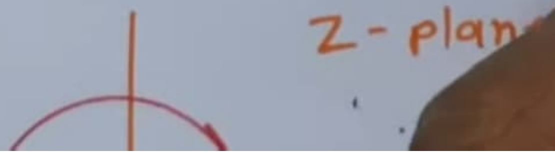
$$\frac{Z}{Z-2} \qquad\qquad \frac{Z}{Z-3}$$

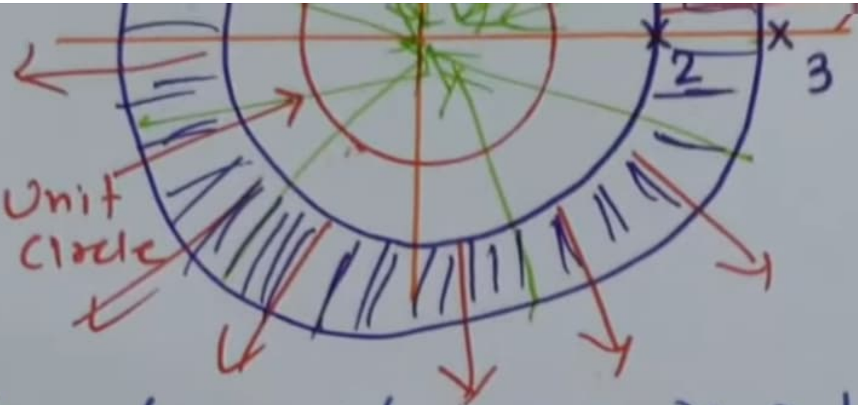
$$\downarrow \qquad\qquad \downarrow$$

$$|Z| > 2 \qquad\qquad |Z| < 3$$

Re $\quad 2 < |Z| < 3$

- If the z-transform of $x(n)$ is ration… …ght sided then ROC is z-plane outside the out…





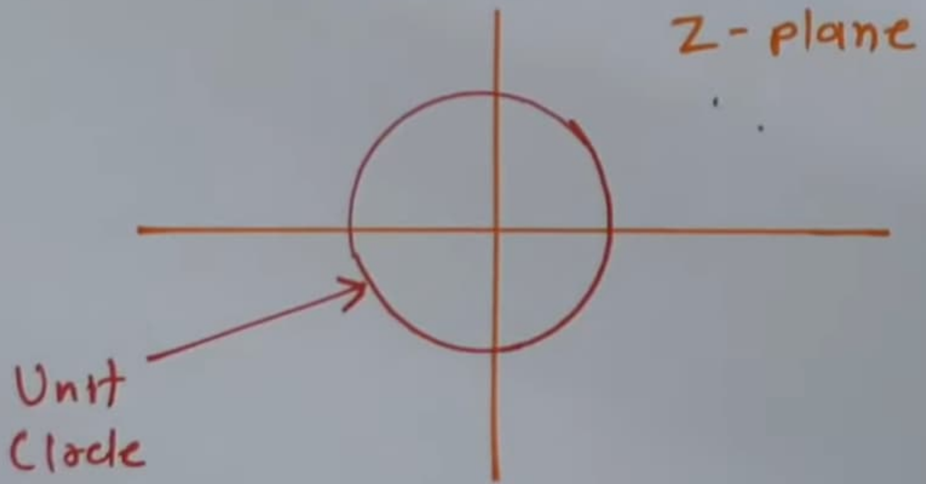
$\frac{1}{z-2}$ $\frac{1}{z-3}$

$\downarrow$ $\downarrow$

$|z| > 2$ $|z| < 3$

ROC: $2 < |z| < 3$

- If the z-transform of $x(n)$ is rational & Right sided then ROC is z-plane outside the outer most pole.



If the z-transform of $x(n)$ is rational & Left sided Then ROC is the region in z-plane inside the inner most pole.